Data Analytics Projects Report

Disclaimer: This is not a report you should modify as per your requirement, bellow i added overview of our project.

Introduction

This report presents a collection of four real-world data analytics projects completed as part of practical learning in data analysis and business intelligence. These projects demonstrate the application of various tools and technologies such as Python, SQL, Power BI, and Excel to solve real-world problems using data.

The objective of these projects is to develop analytical thinking, data processing skills, and the ability to extract meaningful insights for decision-making.

Project 1: Exploratory Data Analysis (EDA) on Cancer Dataset

Objective

To analyze a global cancer dataset and identify patterns, trends, and relationships between various risk factors and cancer severity.

Tools & Technologies

- Python (Pandas, NumPy)
- Matplotlib & Seaborn

Dataset Description

The dataset contains 50,000+ records of cancer patients including:

- Demographics (Age, Gender, Country)
- Risk Factors (Smoking, Alcohol, Obesity, Pollution)
- Clinical Data (Cancer Type, Stage, Treatment Cost)
- Outcomes (Survival Years, Severity Score)

Key Tasks Performed

- Data Cleaning and Preprocessing
- Handling missing values
- Data visualization
- Correlation analysis

Key Insights

- Higher smoking and pollution levels are associated with increased cancer severity
- Obesity and genetic risk also contribute significantly
- Survival years decrease as cancer stage increases

Conclusion

EDA helped uncover critical relationships between lifestyle factors and cancer outcomes, highlighting the importance of preventive healthcare.

Project 2: SQL Analysis on AdventureWorks Dataset

Objective

To analyze business data and answer real-world questions related to sales, customers, and products using SQL.

Tools & Technologies

- SQL Server

Key Tasks Performed

- Writing complex SQL queries
- Using JOINs to combine multiple tables
- Applying aggregate functions (SUM, AVG, COUNT)
- Implementing Window Functions (ROW_NUMBER, RANK)
- Using Subqueries

Business Problems Solved

- Identify top customers based on revenue
- Analyze monthly and yearly sales trends
- Find best-performing products
- Evaluate regional sales performance

Conclusion

This project strengthened SQL skills and demonstrated how databases are used to derive business insights.

Project 3: Live API Data Integration with Python & Power BI

Objective

To build a real-time data pipeline and dashboard using live API data.

Tools & Technologies

- Python (Requests, JSON)
- Power BI

Workflow

1. Fetch real-time data from public APIs
2. Process and clean data using Python
3. Load data into Power BI
4. Create live dashboards

Key Features

- Automated data fetching
- Real-time dashboard updates
- No manual data entry required

Key Insights

- Real-time monitoring of global indicators
- Comparison across countries and regions
- Trend analysis over time

Conclusion

This project demonstrates real-world data engineering and dashboarding skills, making reporting efficient and automated.

Project 4: Company Performance Dashboard in Excel

Objective

To design an interactive dashboard for analyzing company performance.

Tools & Technologies

- Microsoft Excel
- Pivot Tables
- Charts & Graphs

Key Features

- KPIs: Total Sales, Profit, Customers
- Category-wise and region-wise analysis
- Interactive filters (Slicers)
- Time-based trend analysis

Key Insights

- Identification of top-performing categories
- Sales trends over time
- Contribution of different regions to revenue

Conclusion

This project demonstrates how Excel can be used as a powerful business intelligence tool for decision-making.

Overall Learning Outcomes

Through these projects, the following skills were developed:

- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- SQL Query Optimization
- Dashboard Development

- Real-time Data Integration
- Business Insight Generation

Final Conclusion

These projects collectively provide hands-on experience in solving real-world data problems using industry-standard tools. They showcase the ability to work with structured and unstructured data, perform analysis, and present insights through dashboards.

This practical exposure prepares students for roles such as:

- Data Analyst
- Business Analyst
- BI Developer